



Gas Type Classification Using Machine Learning

A Comparative Machine Learning Study

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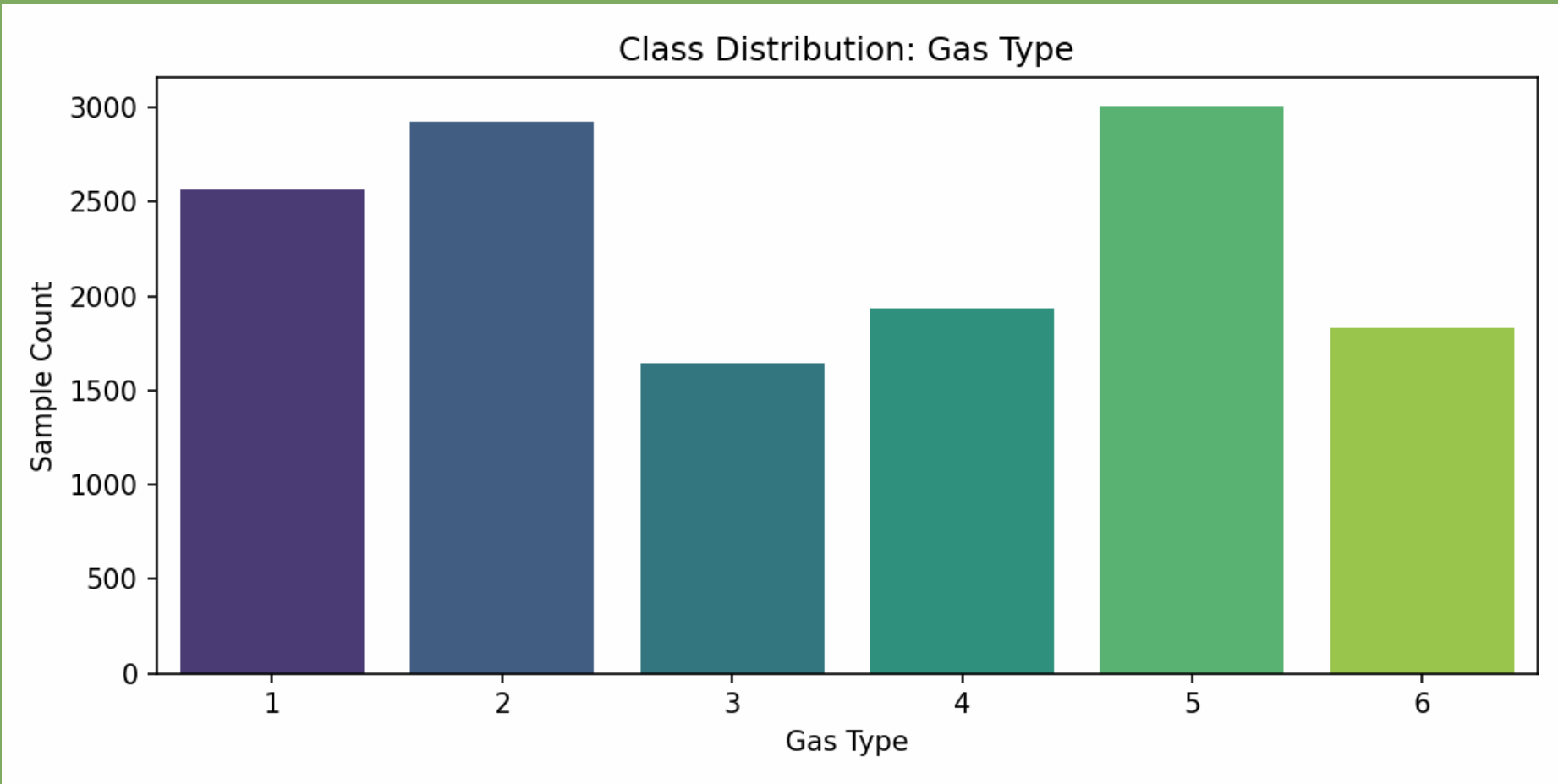
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Introduction:

Chemical gas sensors degrade over time, a phenomenon known as “sensor drift”, making reliable gas identification increasingly difficult without recalibration. This project applies machine learning to classify six distinct gases from sensor array readings collected over 36 months.

- **Dataset:** Gas Sensor Array Drift Dataset (UCI, ID:224)
- 13,910 measurements from 16 metal-oxide sensors
- 6 gas types: Ethanol, Ethylene, Ammonia, Acetaldehyde, Acetone, Toluene
- 128 engineered features per measurement



- *Ethanol-class 1*
- *Ethylene-class 2*
- *Ammonia-class 3*
- *Acetaldehyde-class 4*
- *Acetone-class 5*
- *Toluene-class 6*

Research Questions:

- Which ML algorithm best classifies gas type from sensor data?
- How do simple models compare to ensemble methods?
- Does PCA preserve performance while reducing dimensionality?

Methodology

- **Preprocessing**
- 80/20 stratified train/test split (11,128 / 2,782 samples)
- Features standardized using StandardScaler (fit on training set only)
- PCA reduced 128 features → 12 components, retaining 95% of variance

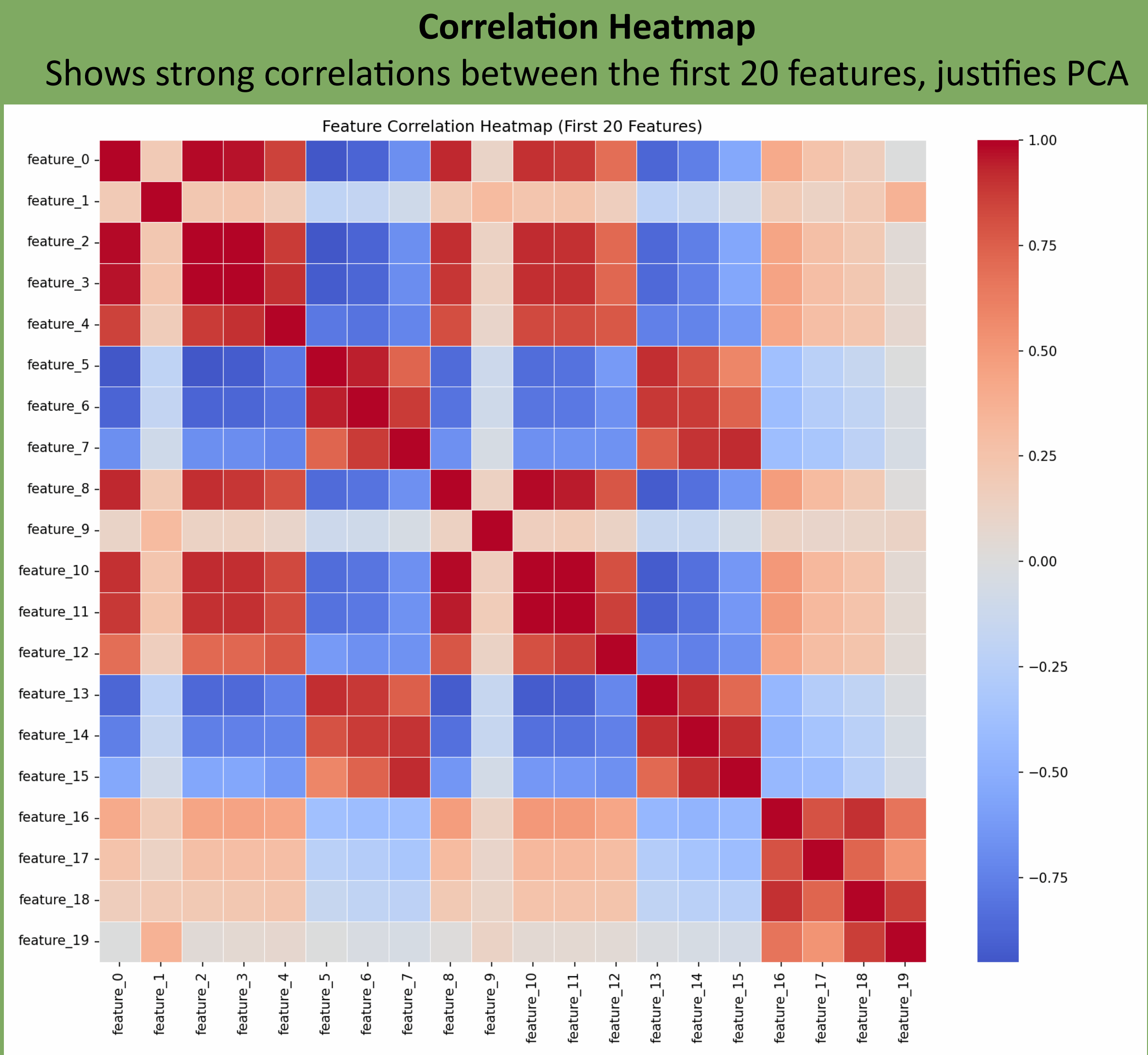
Models Trained:

Model	Type
K-Nearest Neighbors (KNN)	Distance-Based
Decision Tree	Single Tree
Support Vector Machine (SVM)	Kernel-Based
Logistic Regression	Linear Baseline
Random Forest	Ensemble (Bagging)
XGBoost	Ensemble (Boosting)

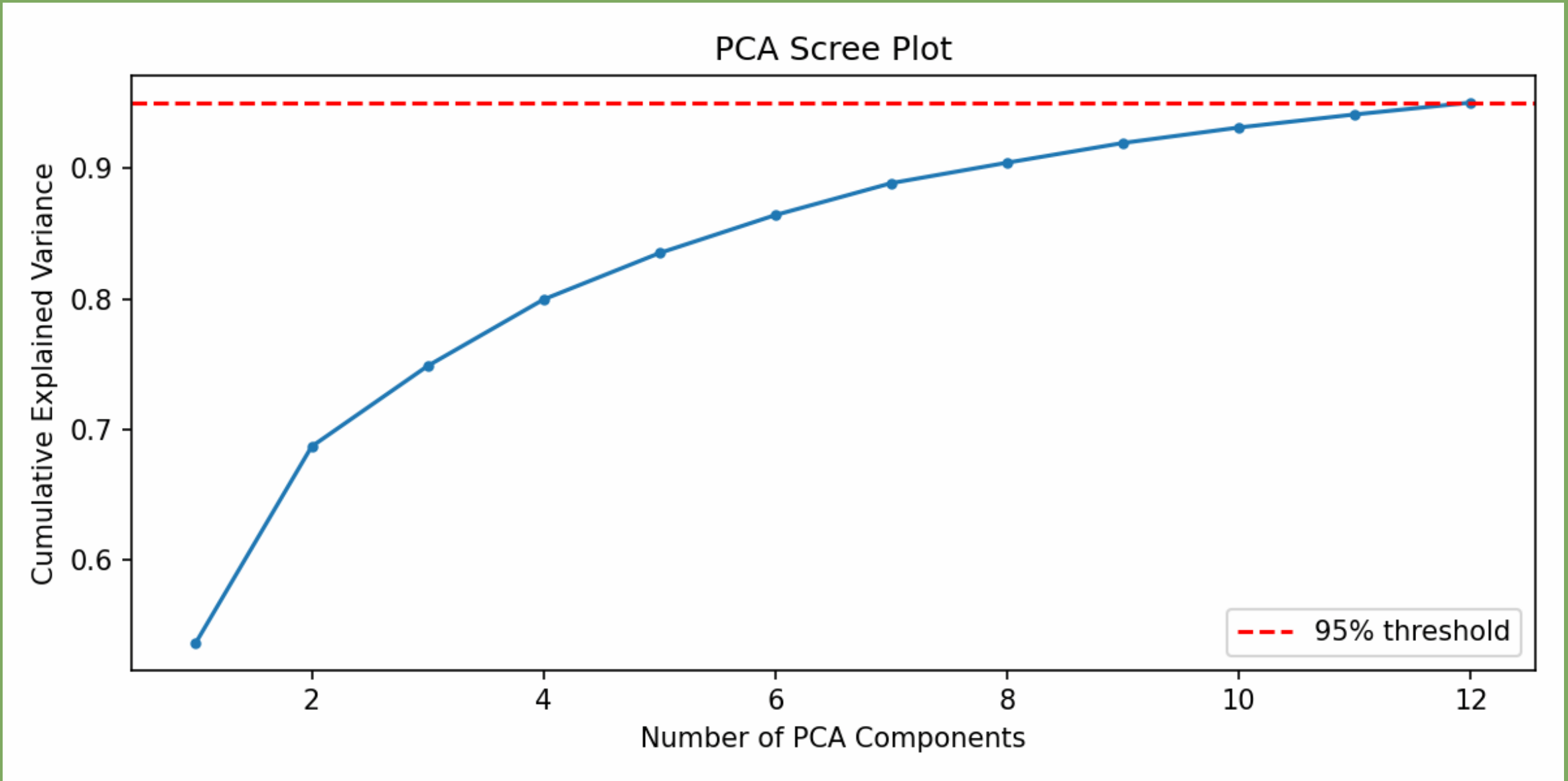
Tuning and Evaluation

- All models tuned via GridSearchCV with 5-fold cross-validation
- Optimized for weighted F1-score
- Evaluated on held-out test set using: Accuracy, Weighted F1, Precision, Recall

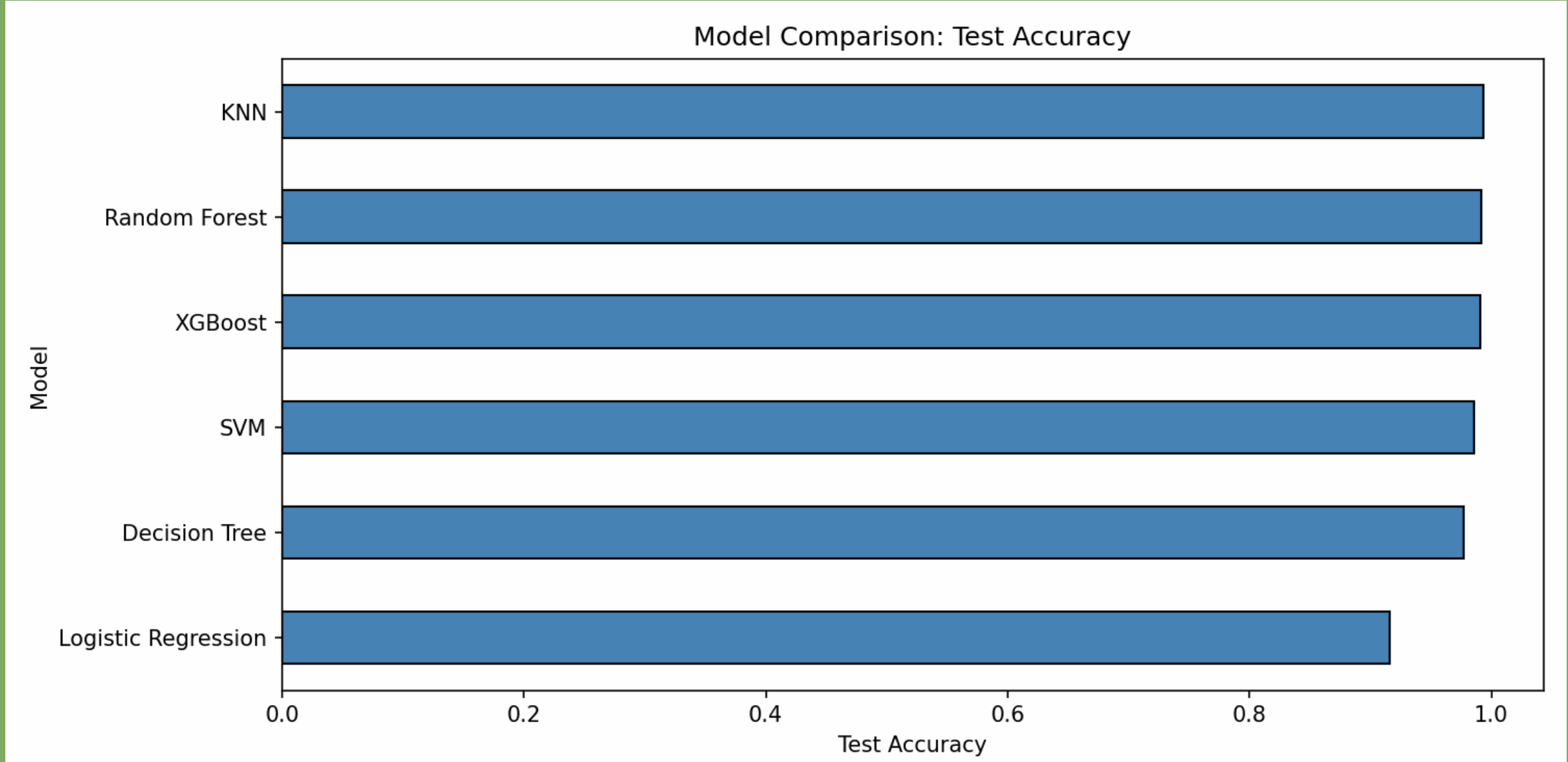
Key Data Visualization:



PCA Scree Plot
Shows how 12 components capture 95% of variance, justifies dimensionality reduction



Model Comparison Bar Chart
Horizontal bars comparing test accuracy across all 6 models at a glance



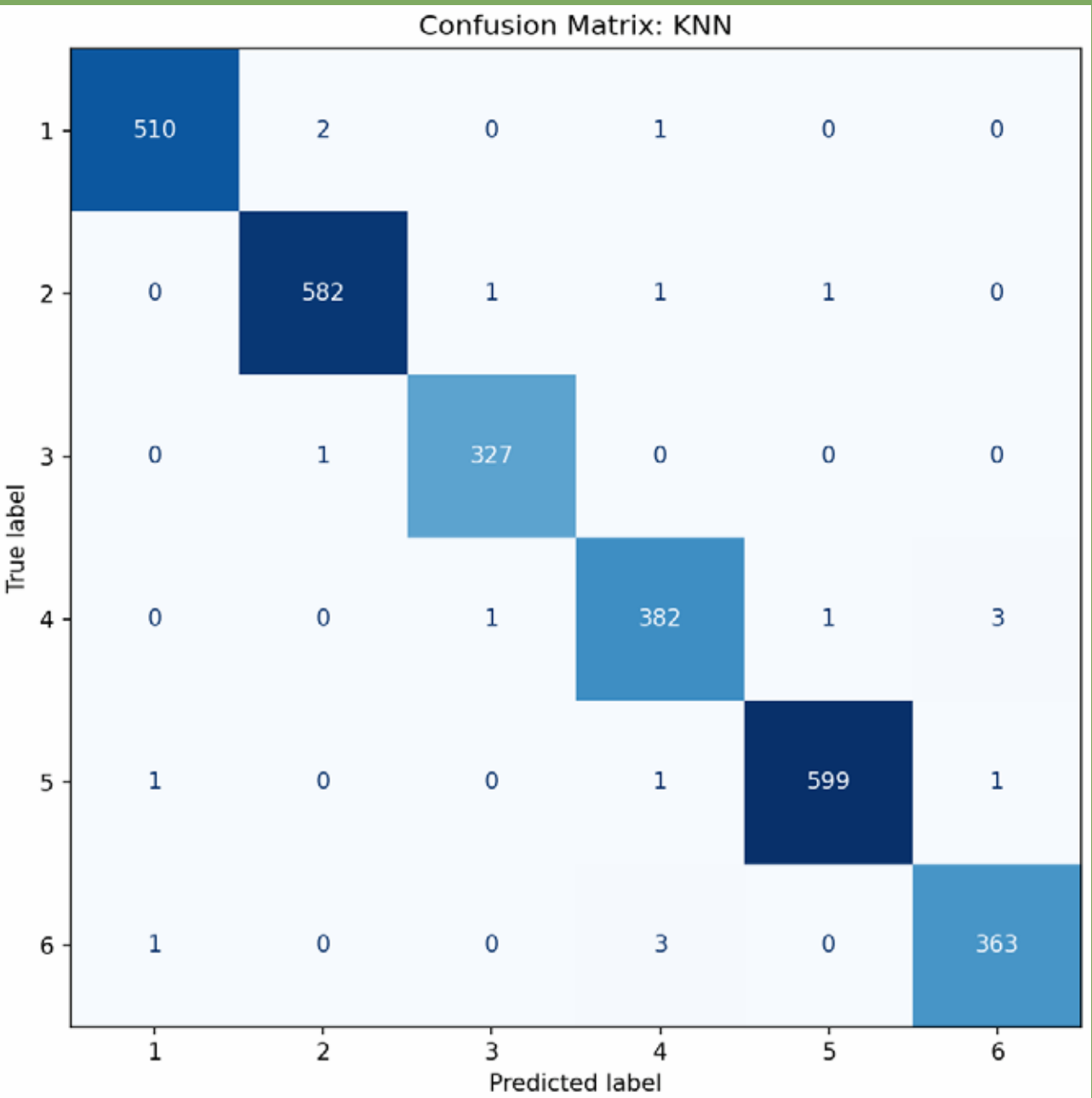
Results:

All six models were evaluated on a held-out test set. Five of six exceeded 98% test accuracy.

Model	Test Accuracy	F1 (Weighted)
KNN	99.32%	0.9932
Random Forest	99.14%	0.9914
XGBoost	99.07%	0.9907
SVM	98.53%	0.9853
Decision Tree	97.70%	0.9770
Logistic Regression	91.55%	0.9151

Best Model: KNN

- Manhattan distance, k=3, distance weighting
- Nearly perfect classification across all 6 gas types
- Primary confusion: Acetaldehyde <-> Toluene



Conclusion

- KNN outperformed all models including ensemble methods, benefiting from PCA's noise reduction and well-separated class boundaries
- PCA reduced features by 90% (128 -> 12) while retaining 95% of variance with no meaningful loss in performance
- Linear models (Logistic Regression) struggled, confirming the classes are not linearly separable

Future Work

- Train on early batches, test on later batches to simulate real sensor drift
- Explore raw time-series features or autoencoder representations

References

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